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#University Category

QS WORLD UNIVERSITY RANKINGS

NO.1 PVT. UNIVERSITY IN ACADEMIC REPUTATION IN INDIA

ACCREDITED WITH GRADE A NAAC

ACCREDITED **GRADE 'A'** BY NAAC

E-LEAD

PERFECT SCORE OF **150/150** AS A TESTAMENT TO EXECEPTIONAL E-LEARNING METHODS

Unit 1 : Introduction to AI

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What is Intelligence

- Intelligence refers to the ability to learn, reason, and solve problems.

Or

- The ability to solve complex problems or make decisions with outcomes benefiting for others.

- For human/animals, problem-solving and decision-making are functions of their nervous systems, including the brain, so intelligence is closely related to the nervous (neural) system.

What is Artificial Intelligence (AI)

- AI is a branch of computer science that focuses on creating systems or machines capable of performing tasks that typically require human intelligence.
- These tasks include:
  - Problem-solving
  - decision-making
  - Learning
  - Perception
  - Language understanding
- AI aims to mimic human cognitive functions through the development of algorithms and computational models.

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Examples of AI

Here are some common examples of AI in everyday life:

1. **Virtual Assistants:** Siri, Alexa, and Google Assistant use AI to understand and respond to voice commands, manage tasks, and provide information.
2. **Recommendation Systems:** Platforms like Netflix and Amazon use AI to recommend movies, shows, and products based on your browsing and purchasing history.
3. **Autonomous Vehicles:** Self-driving cars use AI to navigate roads, recognize obstacles, and make driving decisions.
4. **Chatbots:** Many websites use AI-driven chatbots to provide customer support and answer frequently asked questions.
5. **Image and Speech Recognition:** AI powers tools that can identify objects in images or transcribe spoken language into text.

Examples of AI

6. Fraud Detection: Banks and financial institutions use AI to monitor transactions and detect fraudulent activity by recognizing unusual patterns.
7. Language Translation: Services like Google Translate use AI to translate text and speech between different languages.
8. Personalized Learning: Educational platforms use AI to adapt learning materials and exercises based on a student's progress and needs.
9. Healthcare Diagnostics: AI assists in diagnosing medical conditions by analyzing medical images, patient data, and historical health records.
10. Smart Home Devices: AI in smart home devices like thermostats and security systems learns from user behavior to optimize energy use and enhance security.

Key Components of AI

The key components of AI include:

1. **Data:** AI systems rely on large volumes of data to learn and make decisions. This data can come from various sources, including sensors, databases, and user interactions.
2. **Algorithms:** These are the mathematical and computational procedures used to process data and make predictions or decisions. Common algorithms include machine learning algorithms, optimization techniques, and neural networks.
3. **Models:** AI models are created using algorithms and trained on data. They represent patterns and relationships in the data, enabling the system to make predictions or perform tasks based on new inputs.
4. **Training:** This involves feeding data into an AI model and adjusting the model's parameters to improve its performance. Training helps the model learn from the data and refine its predictions or decisions.
5. **Inference:** After training, AI models are used to make predictions or decisions based on new, unseen data. This process is known as inference and involves applying the learned patterns to real-world scenarios.

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Key Components of AI

6. Hardware: AI systems often require specialized hardware, such as GPUs (Graphics Processing Units) or TPUs (Tensor Processing Units), to handle the large computational demands of training and inference.

7. Software: AI software includes frameworks and libraries that facilitate the development, training, and deployment of AI models. Popular examples are TensorFlow, PyTorch, and Scikit-learn.

8. Evaluation: AI systems need to be evaluated to ensure they are performing accurately and reliably. Evaluation metrics help assess the model's performance and identify areas for improvement.

9. Deployment: Once trained and evaluated, AI models are deployed into production environments where they can interact with users and perform their intended tasks.

10. Feedback Mechanisms: These are used to continually improve the AI system by collecting user feedback and updating the model based on new data and insights.

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3 stages of AI

Narrow AI

Dedicated to assist with or take over specific tasks

General AI

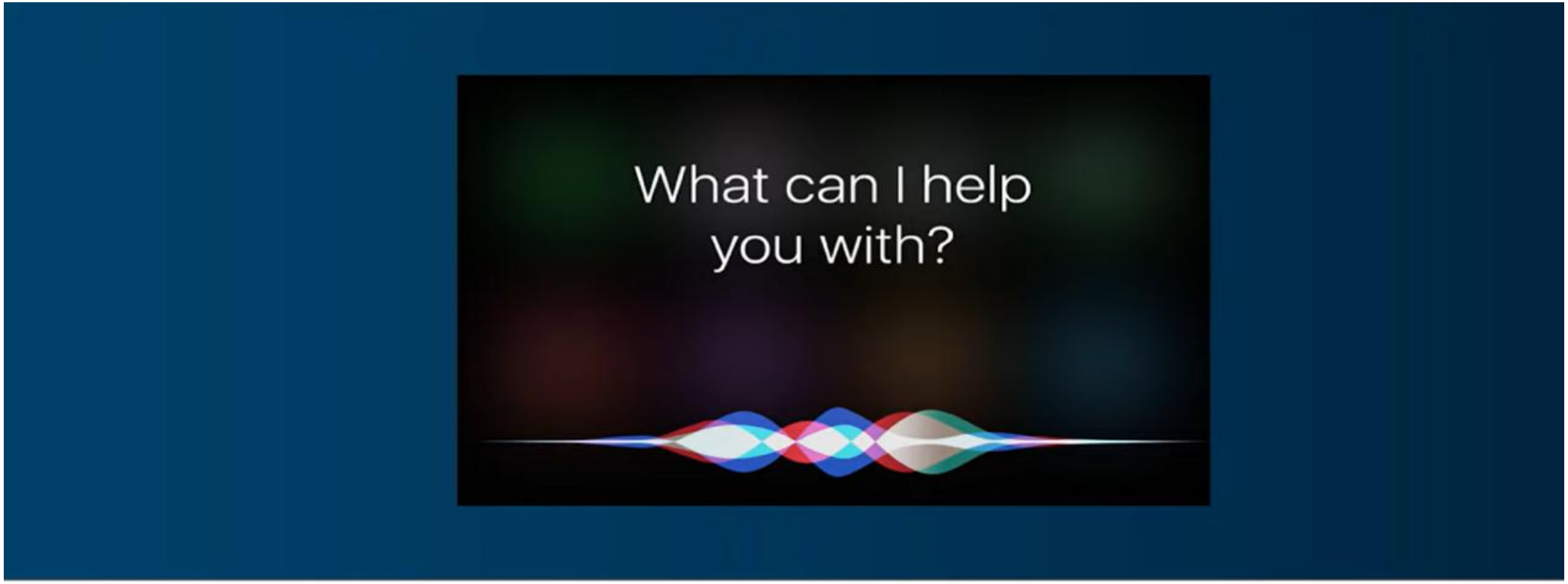
Takes knowledge from one domain, transfers to other domain

Super AI

Machines that are an order of magnitude smarter than humans

Types/Stages of AI

- **Narrow AI (Weak AI)**:
  - **Definition**: AI that is designed and trained for a specific task or a narrow range of tasks.
  - **Example**: Virtual assistants like Siri and Alexa that can perform tasks like setting reminders or playing music.
- **General AI (Strong AI)**:
  - **Definition**: AI with the ability to understand, learn, and apply knowledge across a broad range of tasks, much like a human.
  - **Example**: This level of AI remains theoretical and has not yet been achieved.
- **Superintelligent AI**:
  - **Definition**: AI that surpasses human intelligence and capabilities in all aspects.
  - **Example**: A hypothetical concept that raises ethical and safety concerns.



Artificial Narrow Intelligence (ANI)

Also known as Weak AI, ANI is the stage of Artificial Intelligence involving machines that can perform only a narrowly defined set of specific tasks.

- **AI that is designed and trained for a specific task or a narrow range of tasks.**
- **Example: Virtual assistants that can perform tasks like setting reminders or playing music.**

Artificial General Intelligence (AGI)

Also known as Strong AI, AGI is the stage in the evolution of Artificial Intelligence wherein machines will possess the ability to think and make decisions just like us humans.

- **AI with the ability to understand, learn, and apply knowledge across a broad range of tasks, much like a human.**
- **Example: Smart cars, games, chatbots, etc.**

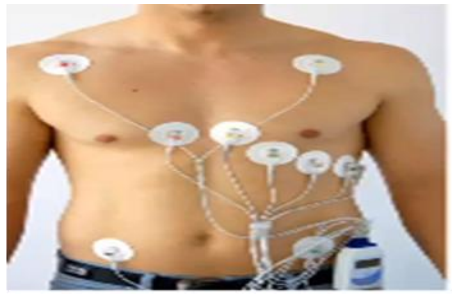
Artificial Super Intelligence (ASI)

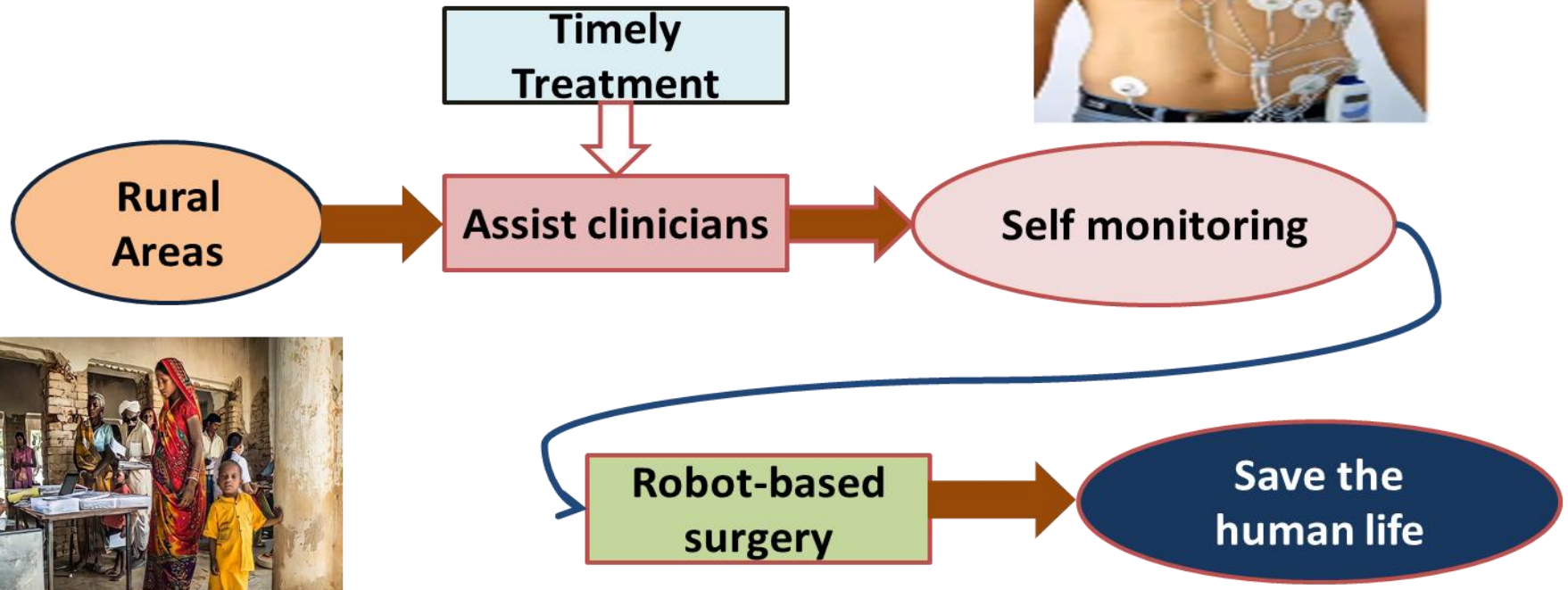
Artificial Super Intelligence is the stage of Artificial Intelligence when the capability of computers will surpass human beings.

- **AI that surpasses human intelligence and capabilities in all aspects.**
- **Example: A hypothetical concept that raises ethical and safety concerns.**

Applications of AI

1. Medical / Healthcare field





I. **Diagnostics:** AI assists in analyzing medical images (e.g., X-rays, MRIs) to detect diseases such as cancer or neurological conditions.

II. **Personalized Medicine:** AI analyzes patient data to tailor treatments and drug recommendations based on individual characteristics.

III. **Drug Discovery:** AI accelerates the process of discovering new drugs by predicting how different compounds will interact.

IV. A medical clinic can use AI systems to organize bed schedules, make a staff rotation and provide medical information.

V. It also has an application in Image guided surgery and image analysis and enhancement.

2. Finance

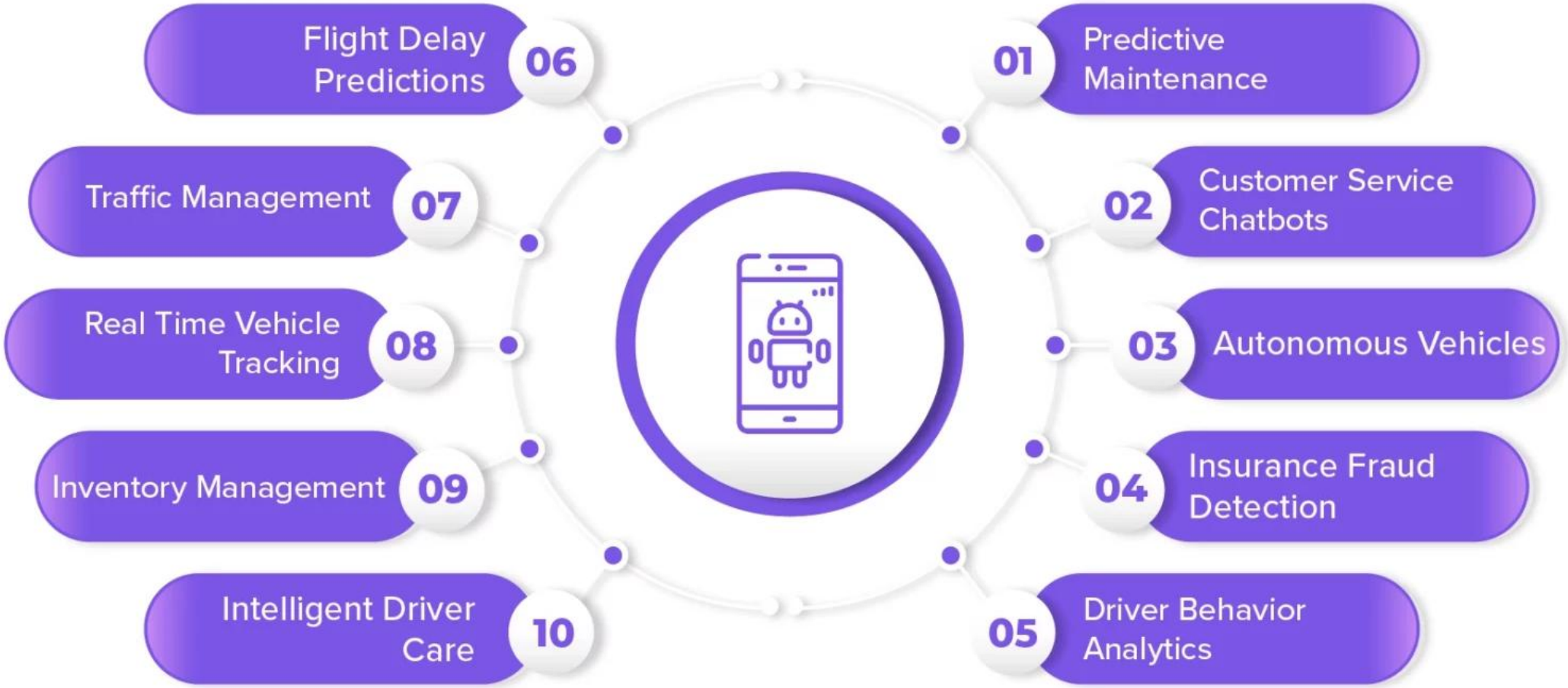
Application of AI in Financial Services
1
Automated Investment Adviced
2
Fraud Detection
3
Risk Management
4
Credit Scoring
5
Personal Financial Management
6
Trading and Execution

I. **Fraud Detection:** AI algorithms analyze transaction patterns to identify and prevent fraudulent activities in real time.

II. **Trading:** AI-driven trading systems can process vast amounts of data faster than humans, identifying patterns and making trades at optimal times. Quantitative trading firms extensively use such technologies.

III. **Customer Service:** Chatbots and virtual assistants handle customer inquiries and provide financial advice, improving customer service efficiency.

3. Transportation

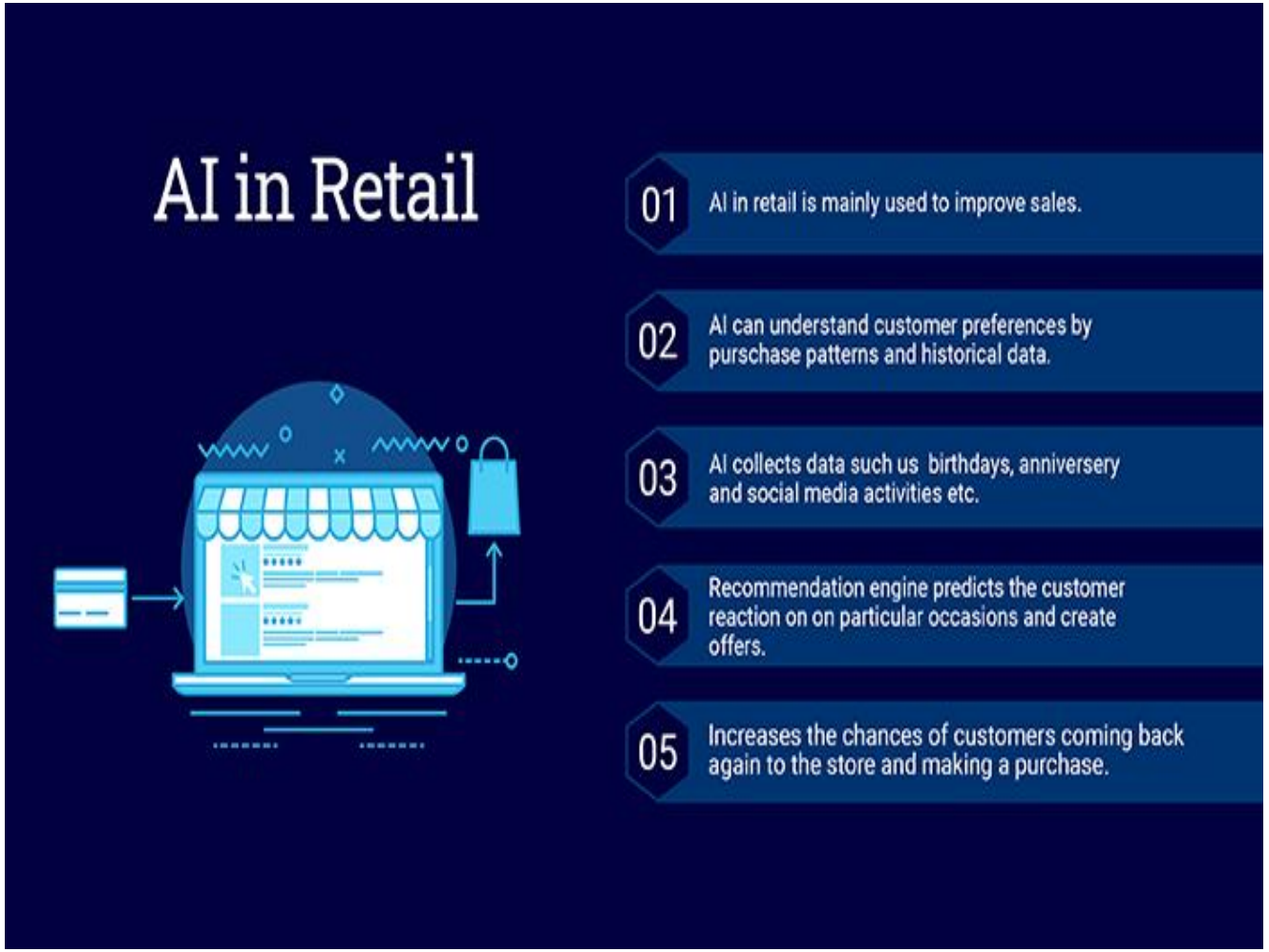


I. **Autonomous Vehicles:** Companies like Tesla and Waymo are developing self-driving cars that use AI to navigate roads, avoid obstacles, and make real-time driving decisions.

II. **Traffic Management:** AI systems optimize traffic flow in cities by analyzing data from various sources, reducing congestion and improving transportation efficiency.

III. **Logistics and Supply Chain:** AI helps in route planning, demand forecasting, and inventory management, enhancing the efficiency of supply chains.

4. Retail


AI in Retail
01 AI in retail is mainly used to improve sales.
02 AI can understand customer preferences by purschase patterns and historical data.
03 AI collects data such us birthdays, anniversery and social media activities etc.
04 Recommendation engine predicts the customer reaction on on particular occasions and create offers.
05 Increases the chances of customers coming back again to the store and making a purchase.

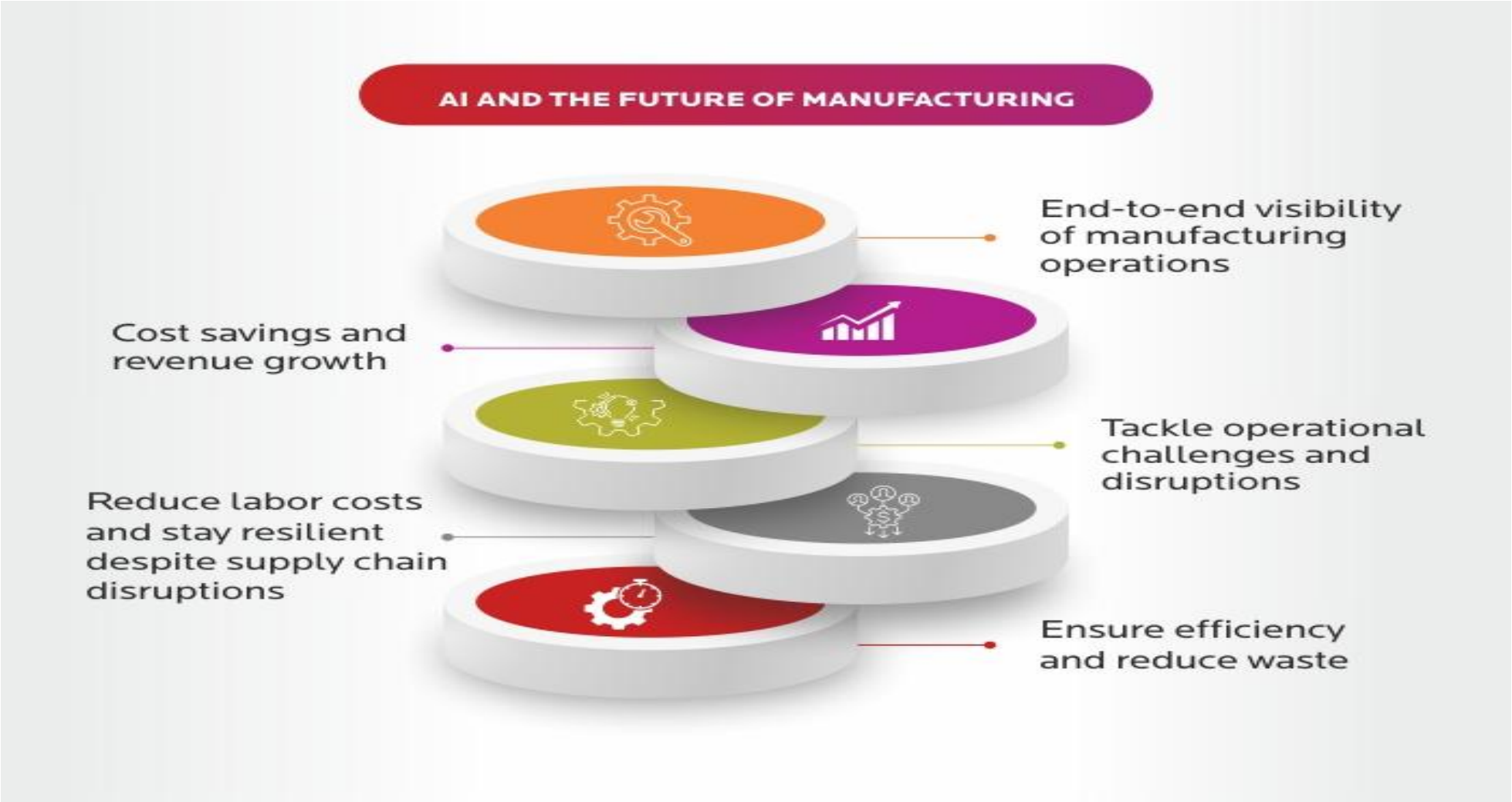
SUPPORT
BUY NOW
TRACK YOUR ORDER
STORE
online store
ONLINE AUCTION
SALE
SOLD
YOU ORDER
APPLICATION & BOOKS STORE

I. **Personalized Shopping Experiences:** AI algorithms analyze customer data to provide personalized product recommendations, improving the shopping experience.

II. **Inventory Management:** AI systems predict demand trends and manage inventory levels to ensure that products are available when needed without overstocking.

III. **Customer Service:** AI-powered chatbots and virtual assistants handle customer inquiries, process orders, and provide support, improving customer satisfaction.

5. Manufacturing

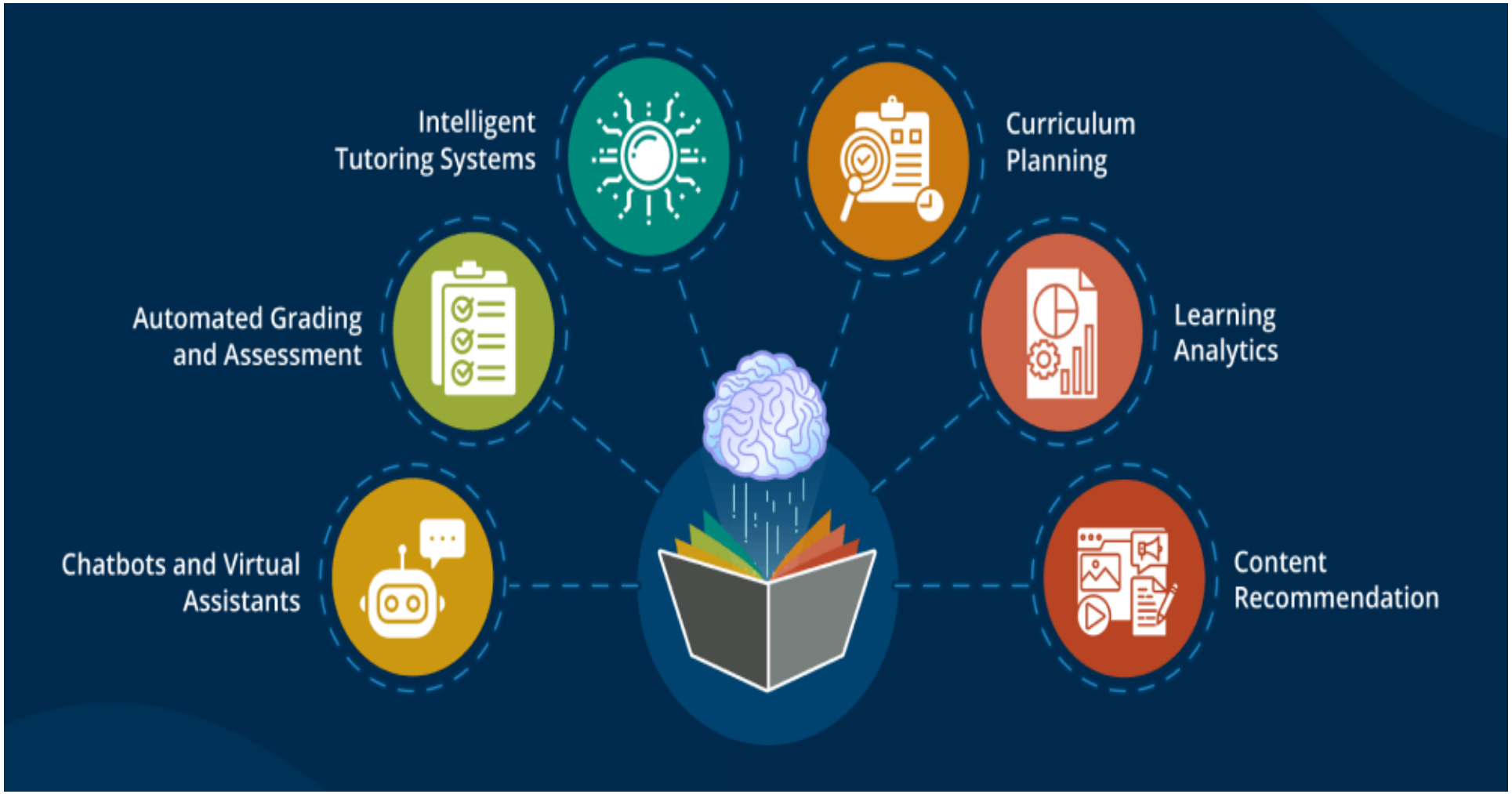


I. **Predictive Maintenance:** AI monitors equipment to predict failures before they occur, reducing downtime and maintenance costs.

II. **Quality Control:** AI systems inspect products for defects, ensuring higher quality and consistency in manufacturing processes.

III. **Robotic Process Automation:** AI-driven robots perform repetitive tasks with high precision, increasing production efficiency and safety.

6. Education

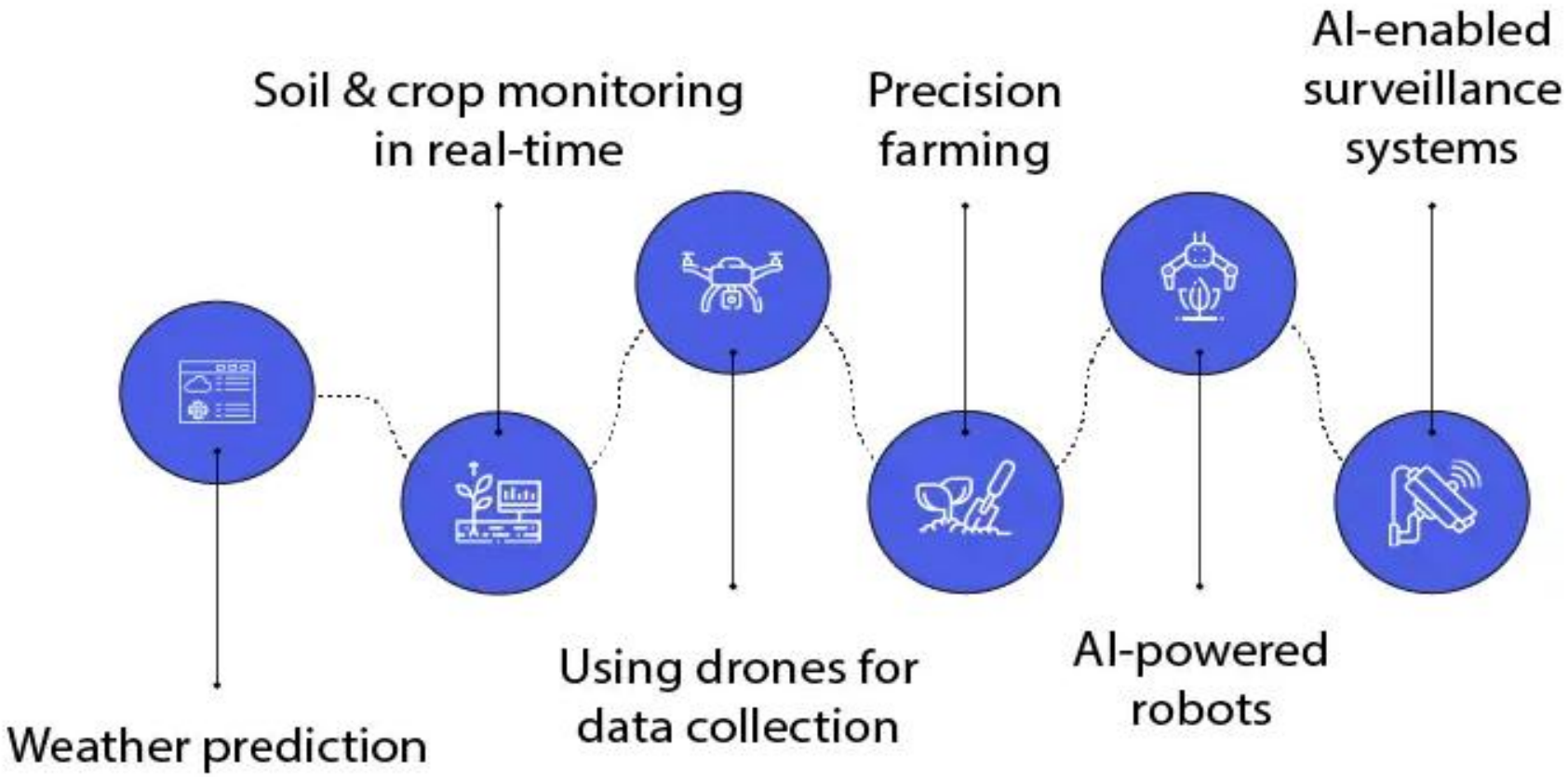

Intelligent Tutoring Systems
Curriculum Planning
Automated Grading and Assessment
Learning Analytics
Chatbots and Virtual Assistants
Content Recommendation

I. **Personalized Learning:** AI provides customized learning experiences based on individual student needs, preferences, and progress.

II. **Administrative Tasks:** AI automates administrative tasks such as grading and scheduling, allowing educators to focus more on teaching.

III. **Virtual Tutors:** AI-powered virtual tutors offer additional support to students, helping them understand complex subjects and providing instant feedback.

7. Agriculture

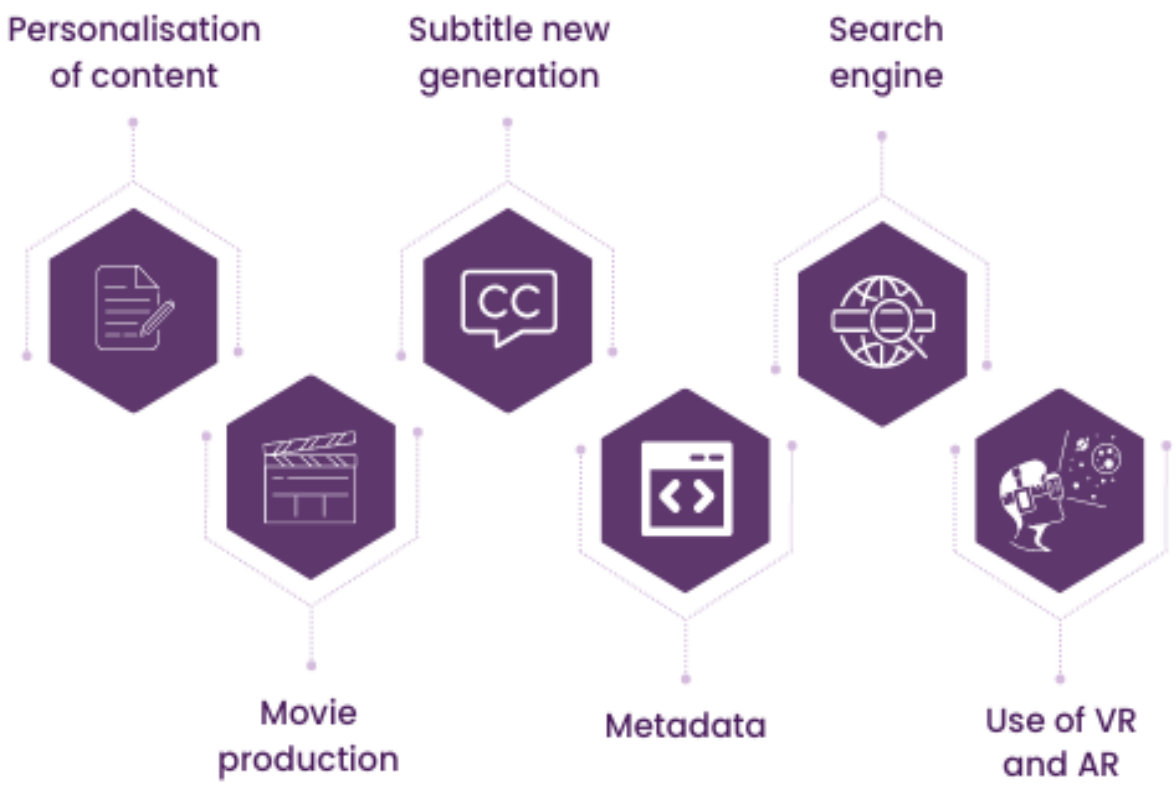


I. **Precision Farming:** AI analyzes data from sensors and drones to optimize planting, watering, and harvesting, increasing crop yields and resource efficiency.

II. **Crop Monitoring:** AI systems detect diseases, pests, and nutrient deficiencies in crops, enabling timely intervention and reducing losses.

III. **Supply Chain Optimization:** AI improves the efficiency of the agricultural supply chain by predicting demand and optimizing logistics.

8. Entertainment



I. **Content Creation:** AI creates music, art, and literature, pushing the boundaries of creativity. Examples include AI-generated music by Amper Music and AI-written articles by tools like GPT-3.

II. **Personalized Content:** Streaming services like Netflix and Spotify use AI to recommend movies, shows, and music based on user preferences and viewing/listening history.

III. **Game Development:** AI enhances video game development by creating intelligent NPCs (non-player characters) and procedural content generation.

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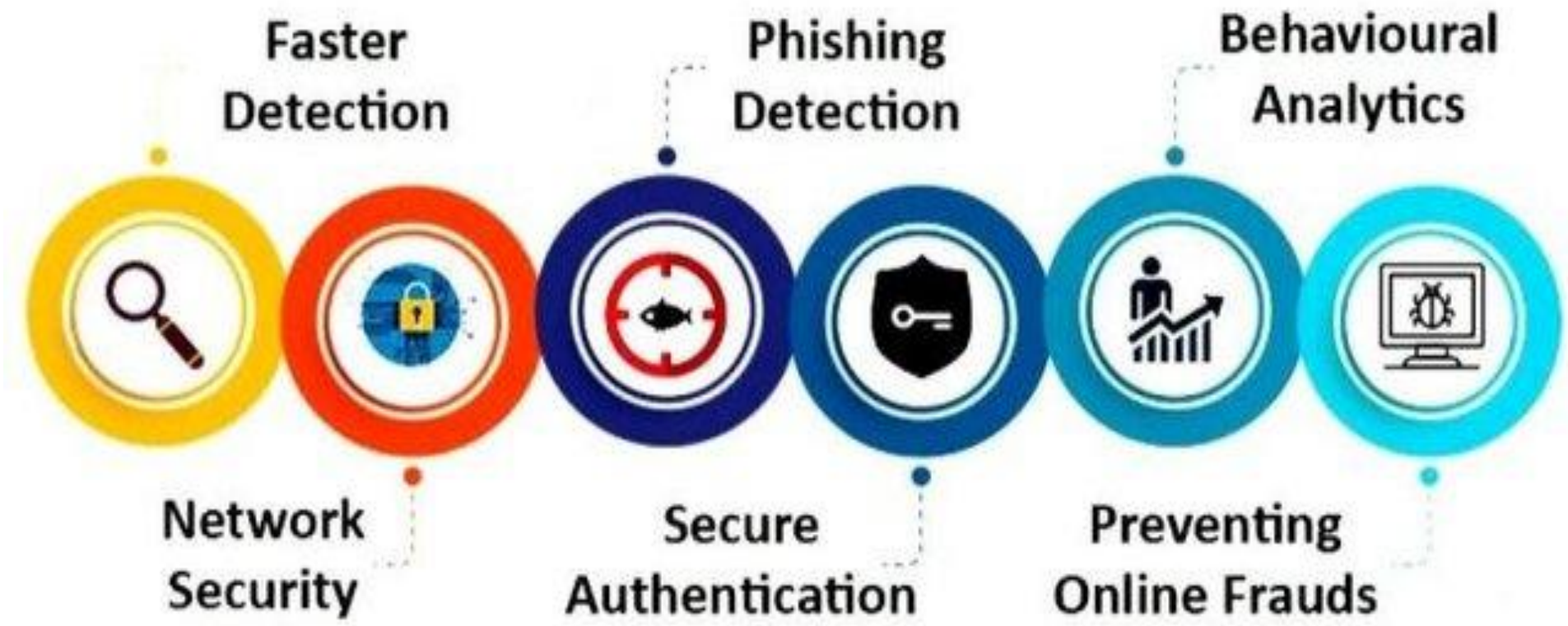
9. Security


Faster Detection
Phishing Detection
Behavioural Analytics
Network Security
Secure Authentication
Preventing Online Frauds

I. **Surveillance:** AI systems analyze video feeds to detect unusual activities and potential security threats in real time.

II. **Cybersecurity:** AI detects and responds to cyber threats by analyzing patterns in network traffic and identifying anomalies.

III. **Identity Verification:** AI enhances security in access control systems through facial recognition and biometric analysis.

10. Environment

I. **Climate Modeling:** AI analyzes climate data to improve weather forecasts and model climate change scenarios.

II. **Wildlife Conservation:** AI tracks animal movements and monitors habitats to aid in wildlife conservation efforts.

III. **Energy Management:** AI optimizes energy use in buildings and industries, contributing to more sustainable energy consumption.

The applications of AI are vast and continually expanding, transforming industries and improving efficiency, accuracy, and outcomes across various fields.

As AI technology evolves, it is expected to play an even more integral role in solving complex global challenges and driving innovation.

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AI Techniques

- AI techniques refer to a set of methods and algorithms used to develop intelligent systems that can perform tasks requiring human-like intelligence.

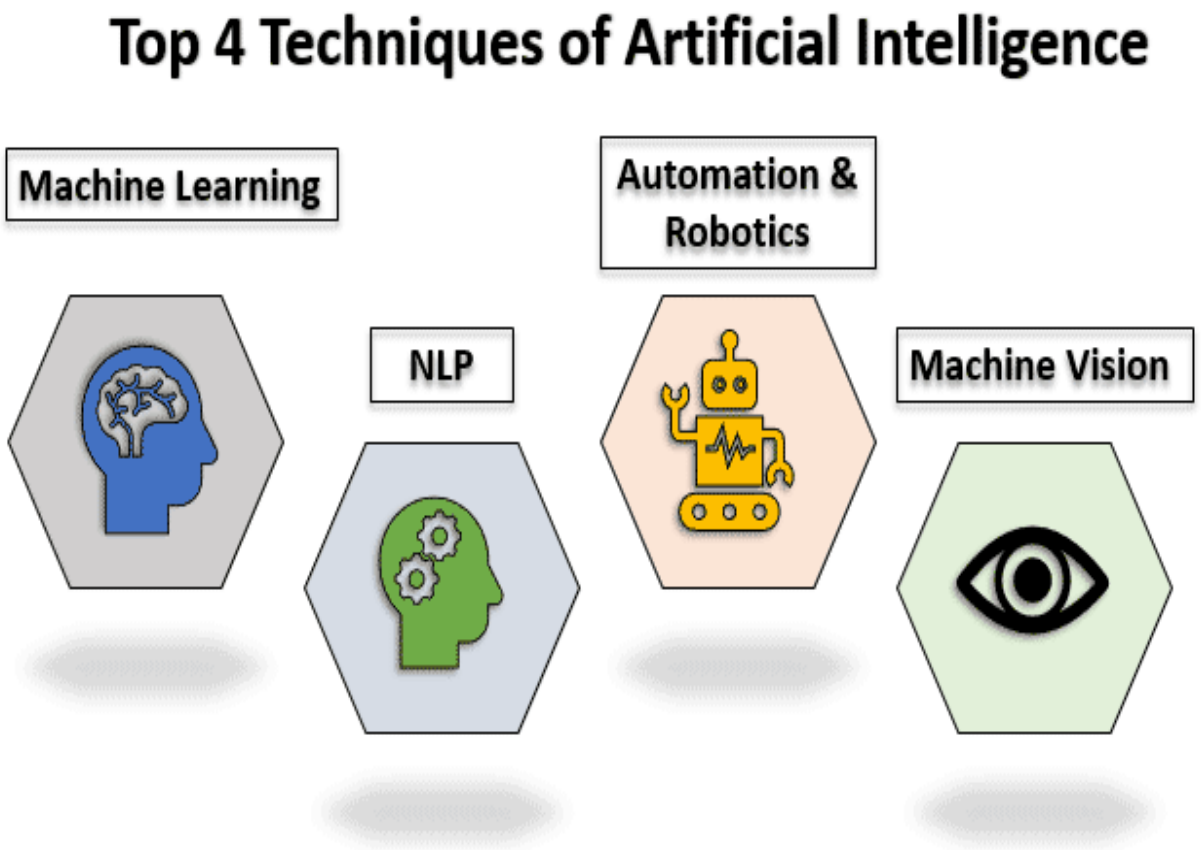
Key Techniques:

Machine Learning

Deep Learning

Natural Language Processing (NLP)

Computer Vision

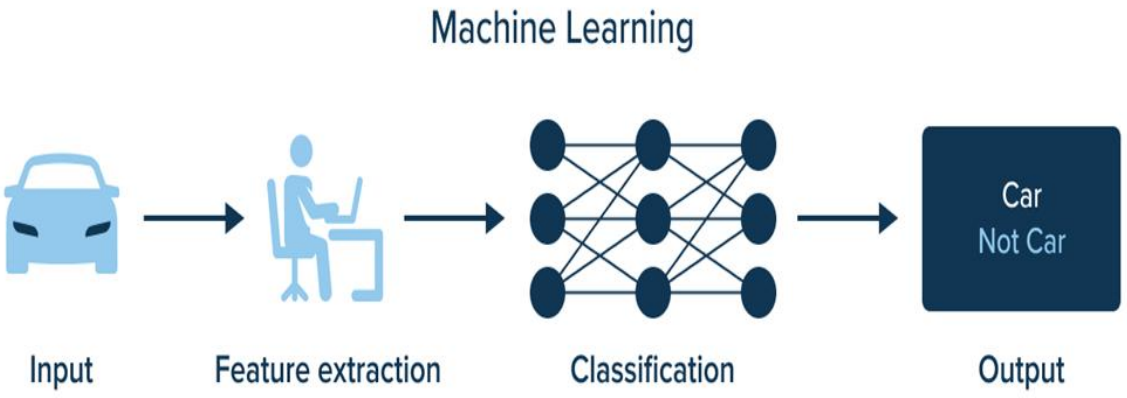


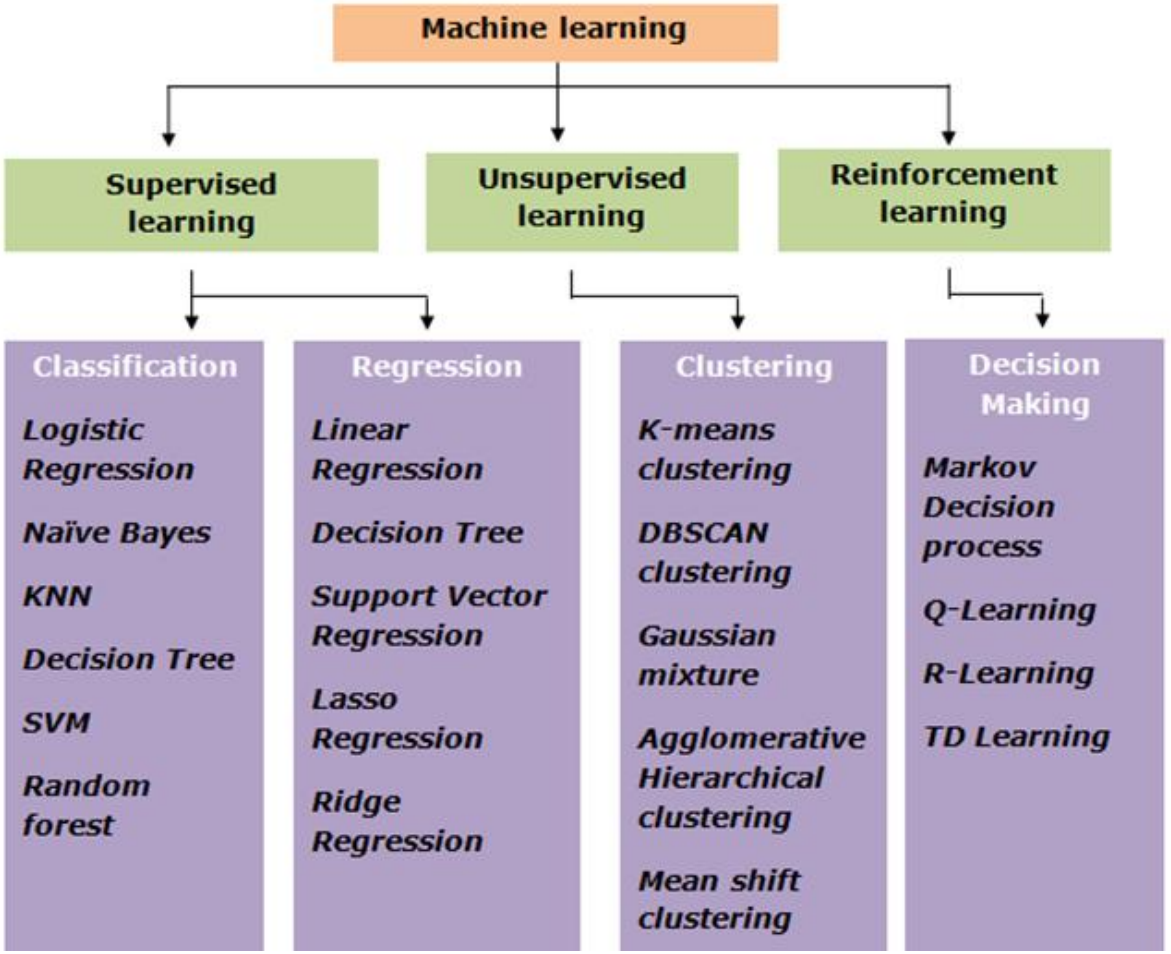
Machine Learning

- Machine learning is a subset of AI techniques that involves training algorithms to learn from data.

Types of ML:

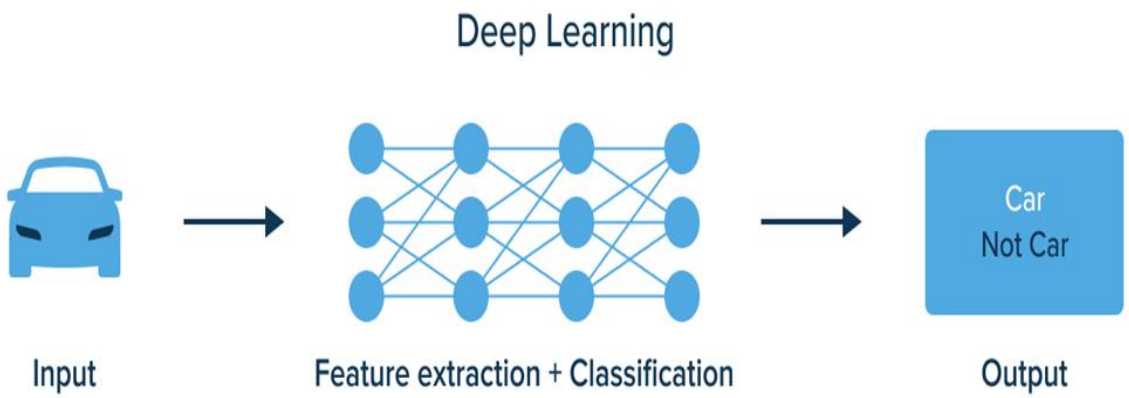
1. Supervised Learning
2. Unsupervised Learning
3. Semi supervised Learning
4. Reinforcement Learning





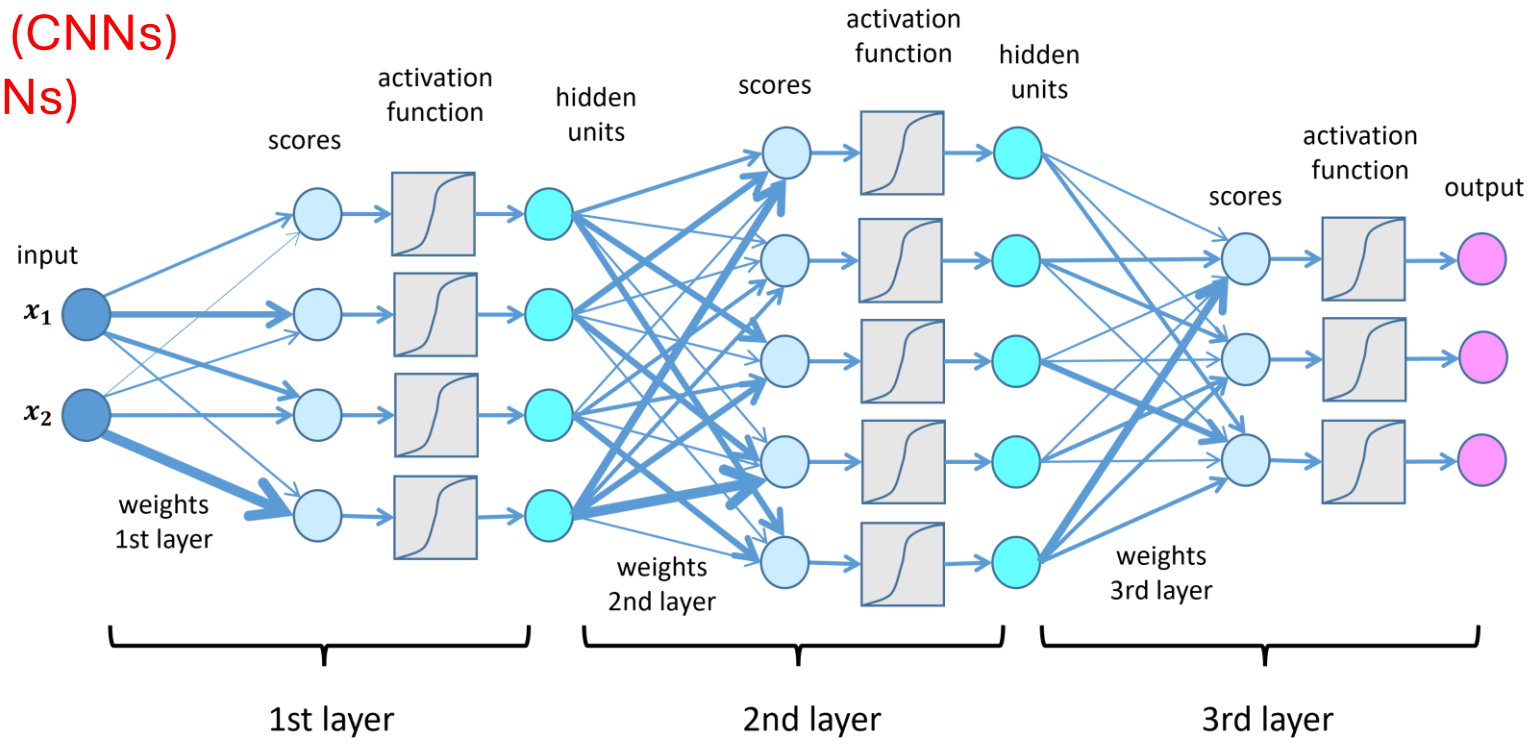
Deep Learning

- Deep learning is a specialized form of machine learning that uses neural networks with multiple hidden layers.



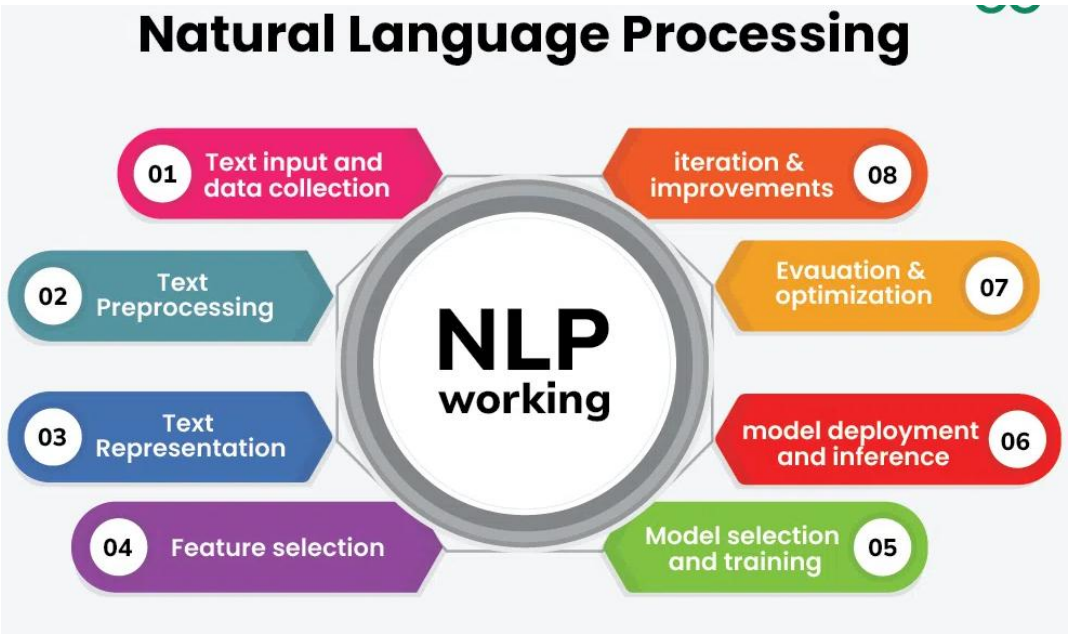
Key Concepts:

1. Neural Networks
2. Convolutional Neural Networks (CNNs)
3. Recurrent Neural Networks (RNNs)



Natural Language Processing (NLP)

- Natural Language Processing involves programming computers to process human languages to facilitate interactions between humans and computers.



Key Techniques:

- Tokenization,
- Sentiment Analysis
- Machine Translation

Applications:

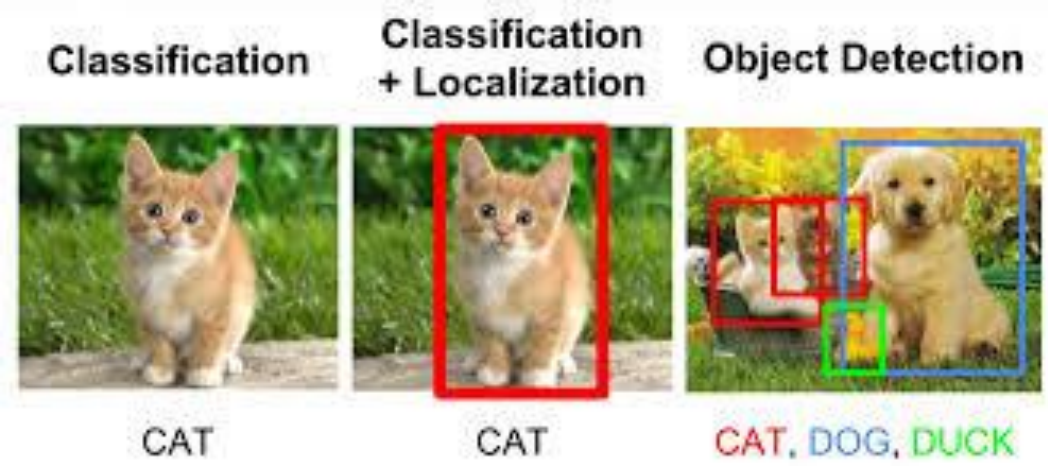
- Chatbots,
- Language Translation
- Text Analytics

Computer Vision

- Computer vision involves teaching machines to interpret and analyze visual information from the real world.

Key Applications:

- Image Classification
- Object Détection
- Image Segmentation.

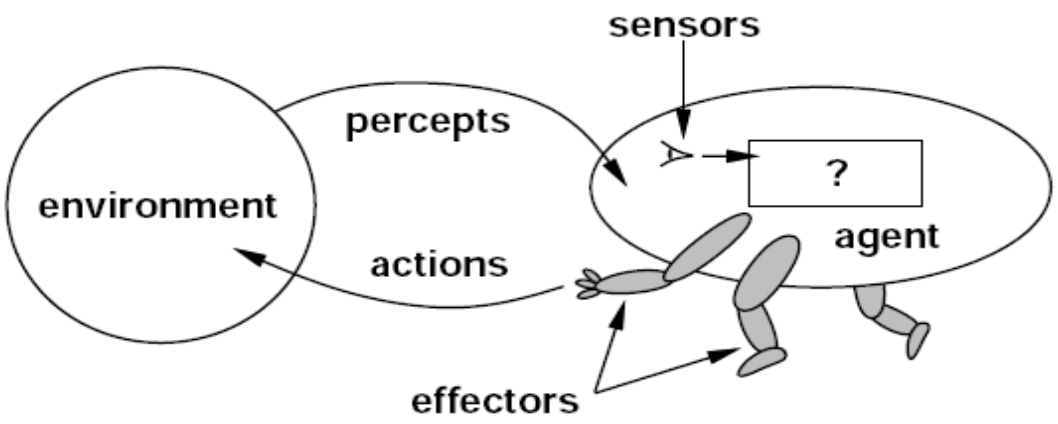


Summary

- AI techniques enable machines to simulate human intelligence.
- Machine learning trains algorithms to learn from data.
- Deep learning uses neural networks with multiple layers.
- NLP enables machines to understand and generate human language.
- Reinforcement learning involves an agent learning by interacting with its environment.
- Genetic algorithms are inspired by natural selection.
- Expert systems emulate human decision-making abilities.
- Fuzzy logic handles uncertainty and imprecision.

Agents/Intelligent Agents

- An **AI Agent** is a system that **perceives its environment, makes decisions, and takes actions** to achieve goals.
- Core cycle: **Perception → Decision → Action → Feedback.**
- Like a human: See → Think → Do → Learn.



➢ Example: A human agent has eyes, ears, and other organs for sensors and hands, legs, vocal tract, and so on for actuators.

Agents/Intelligent Agents

Everyday Examples of AI Agents

- **Virtual Assistants**: Alexa, Siri – respond to voice commands.
- **Game AI**: Enemies adapting to your moves.
- **Navigation**: Google Maps rerouting around traffic.
- **Smart Homes**: Thermostats adjusting temperature automatically.

Real-World Applications

- **Healthcare**: Monitoring patients with wearables.
- **Autonomous Cars**: Self-driving vehicles reacting in real-time.
- **Finance**: AI agents making stock trades.
- **Robotics**: Amazon warehouse robots managing packages.

Agents/Intelligent Agents

- A robotic agent might have cameras and infrared range finders for sensors and various motors for actuators.
- A software agent receives keystrokes, file contents, and network packets as sensory inputs and acts on the environment by displaying on the screen, writing files, and sending network packets.
- The agent operates autonomously, meaning it is not directly controlled by a human operator.
- Agents can be classified into different types **based on their characteristics**, such as
  - reactive or proactive
  - fixed or dynamic environment
  - single or multi-agent systems.

Agents/Intelligent Agents

Category	Type	Description	Example
Behavior	**Reactive Agents**	Respond directly to environment stimuli without long-term planning.	Thermostat, obstacle-avoiding robot
	Proactive Agents	Take initiative, plan ahead, and work towards goals.	GPS route planner, self-driving car
Environment Handling	**Fixed Environment**	Operate in a stable, predictable environment.	Chess AI, assembly-line robot
	Dynamic Environment	Adapt to changing, unpredictable conditions.	Stock trading bot, autonomous drone
Number of Agents	**Single-Agent System**	Works independently in isolation.	Personal voice assistant (Siri, Alexa)
	Multi-Agent System	Multiple agents interact/cooperate/ compete within the same environment.	Swarm robotics, online multiplayer game AI

Rules all AI agents must adhere to:

- **Perception Rule** – An agent must perceive its environment using sensors (e.g., camera, microphone).
- **Decision-Making Rule** – Based on perceptions, it must choose an appropriate action.
- **Action Rule** – Use actuators (motors, speakers, software commands) to act on the environment.
- **Rationality Rule** – The agent should always act rationally, aiming to maximize performance or achieve its goals.
- **Learning Rule** – Agents should improve over time by learning from feedback and past actions.

Functions of AI Agents

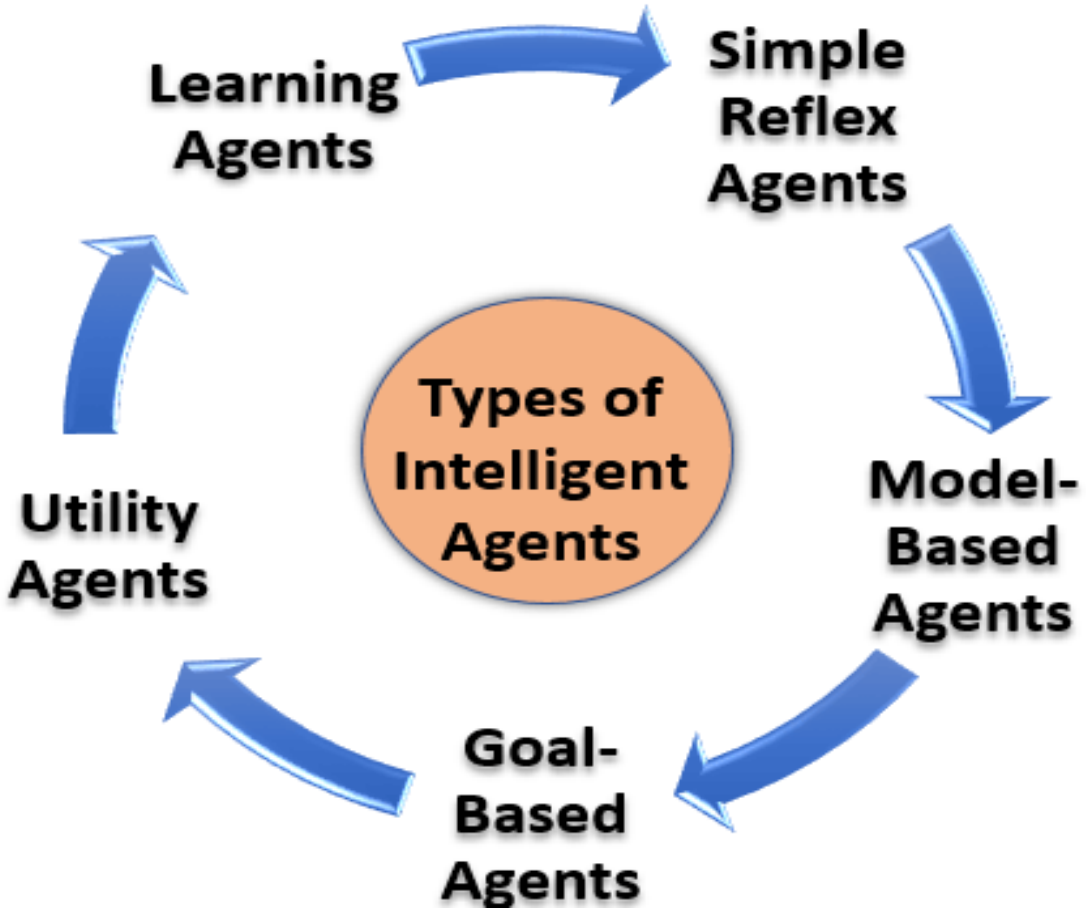
- **Autonomy** – Operate without continuous human intervention.
- **Goal-Oriented Behavior** – Work towards defined goals (e.g., deliver package, answer query).
- **Adaptability** – Adjust actions based on changes in the environment.
- **Interaction** – Communicate with humans or other agents when needed.
- **Optimization** – Find efficient solutions, balancing accuracy, cost, time, and resources.
- **Decision Support** – Assist humans in decision-making (e.g., medical diagnosis systems).
- **Learning & Improvement –**

A voice assistant (Alexa):

➢ Rules → Listen (sensor), process speech (decision), respond with audio (actuator).

Types of Agents in AI:

Study from this link : https://www.geeksforgeeks.org/agents-artificial-intelligence/



Agents/Intelligent Agents

1. **Simple Reflex Agents**

- Work only on condition–action rules ("if this → then that").
- No memory of the past.
- Example: A thermostat – if temperature < 20°C, then turn heater ON.
- Limitation: Can't handle complex scenarios.

2. **Model-Based Reflex Agents**

- Have an internal model of the world to handle partially observable environments.
- They use memory to track what's happening.
- Example: A robot vacuum cleaner remembers where it already cleaned and avoids repeating.

Agents/Intelligent Agents

3. Goal-Based Agents

- Make decisions by considering goals and planning actions to achieve them.
- More flexible than reflex agents.
- Example: A GPS navigation system – goal = reach destination, it calculates best route.

4. Utility-Based Agents

- Not just about reaching a goal, but choosing the best possible action by maximizing a utility function (happiness, efficiency, profit, etc.).
- Example: A self-driving car choosing a route that balances speed, safety, and fuel efficiency.

Agents/Intelligent Agents

5. Learning Agents

- Improve performance over time by learning from past actions and feedback.
- Have four components: learning element, critic, performance element, and problem generator.
- Example: A recommendation system (like Netflix or YouTube) improves suggestions based on your watching habits.

Simple Analogy

1. **Reflex Agent** → Student who answers only from rote memory.
2. **Model-Based** → Student who remembers past lessons while answering.
3. **Goal-Based** → Student aiming to score high by solving carefully.
4. **Utility-Based** → Student balancing high marks **and** finishing exam on time.
5. **Learning Agent** → Student who learns from mistakes and does better in the next test.

Advantages of AI

1. Efficiency and Productivity

- **Automation of Routine Tasks:** AI can automate repetitive tasks, freeing up human workers to focus on more complex and creative activities.
- **24/7 Availability:** AI systems can operate continuously without fatigue, providing services and monitoring systems around the clock.

2. Accuracy and Precision

- **Reduced Errors:** AI systems can process large amounts of data with high accuracy, reducing the likelihood of human errors.
- **Precision in Tasks:** In fields like medicine and manufacturing, AI can perform tasks with a level of precision unattainable by humans.

3. Data Processing and Analysis

- **Big Data Insights:** AI can analyze vast datasets quickly, identifying patterns and trends that would be impossible for humans to discern.
- **Predictive Analytics:** AI systems can forecast future trends and outcomes, aiding in decision-making and strategic planning.

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4. Innovation and Discovery

New Solutions: AI can help discover new solutions to complex problems, such as drug discovery in pharmaceuticals and climate modeling.

Enhanced Creativity: AI tools assist in creative fields by generating music, art, and content, pushing the boundaries of human creativity.

5. Cost Savings

Operational Efficiency: By automating processes and optimizing operations, AI can significantly reduce costs for businesses.

Reduced Labor Costs: AI can perform tasks that would otherwise require a large workforce, leading to labor cost savings.

6. Improved Customer Experience

Personalization: AI enables personalized experiences in retail, entertainment, and online services, increasing customer satisfaction.

Enhanced Support: AI-driven chatbots and virtual assistants provide quick and efficient customer service.

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Disadvantages of AI

1. Job Displacement

- **Unemployment:** The automation of tasks traditionally performed by humans can lead to job losses and increased unemployment in certain sectors.
- **Skill Gaps:** There is a growing need for workers to acquire new skills to stay relevant in an AI-driven job market.

2. High Implementation Costs

- **Initial Investment:** Developing and implementing AI systems can be expensive, requiring significant upfront investment.
- **Maintenance Costs:** AI systems require ongoing maintenance, updates, and fine-tuning, adding to operational costs.

3. Ethical and Privacy Concerns

- **Bias and Discrimination:** AI systems can perpetuate and even exacerbate biases present in the training data, leading to unfair outcomes.
- **Privacy Issues:** The use of AI in surveillance and data analysis raises concerns about privacy and the potential misuse of personal data.

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4. Complexity and Understanding

Lack of Transparency: Many AI systems, especially those using deep learning, operate as "black boxes," making it difficult to understand how they make decisions.

Dependence on Quality Data: AI's effectiveness is heavily dependent on the quality and quantity of data available, and poor data can lead to poor outcomes.

5. Security Risks

Cyber Attacks: AI systems can be vulnerable to hacking and other cyber threats, potentially leading to significant security breaches.

Autonomous Weapons: The development of AI-powered weapons poses significant ethical and security risks.

6. Loss of Human Touch

Impersonal Interactions: Over-reliance on AI for customer service and other interactions can lead to a lack of human touch and personal connection.

Dehumanization: In some applications, the use of AI can lead to the dehumanization of certain processes and interactions.

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Limitations of AI

Artificial Intelligence (AI) has transformative potential, but it also comes with several limitations that impact its effectiveness and broader adoption. Here are some of the primary limitations of AI:

1. Data Dependency

- **Quality of Data:** AI systems require high-quality data for training. Poor-quality data, including biased, incomplete, or noisy data, can lead to inaccurate models and flawed outcomes.
- **Quantity of Data:** Many AI models, especially those using deep learning, need large amounts of data to perform well. In domains where data is scarce, this can be a significant barrier.

2. Lack of Generalization

- **Specific Task Performance:** Most AI systems are designed for specific tasks and cannot generalize their knowledge to different contexts or tasks. They excel in narrow applications but struggle with tasks outside their training scope.
- **Transfer Learning Limitations:** While techniques like transfer learning aim to apply knowledge from one domain to another, they are not always effective and require careful adaptation.

3. Explainability and Transparency

- **Black Box Models:** Many AI models, particularly deep learning networks, operate as "black boxes" with decision-making processes that are difficult to understand or explain.
- **Trust Issues:** The lack of transparency can lead to trust issues among users and stakeholders, especially in critical applications like healthcare and finance where understanding the rationale behind decisions is crucial.

4. Ethical and Bias Concerns

- **Bias in AI:** AI systems can inherit biases present in their training data, leading to discriminatory outcomes in areas like hiring, lending, and law enforcement.
- **Ethical Dilemmas:** AI poses ethical challenges related to privacy, surveillance, and the potential for misuse, such as autonomous weapons or deepfakes.

5. Computational and Resource Intensity

- **High Computational Costs:** Training advanced AI models, particularly deep neural networks, requires significant computational resources, which can be expensive and environmentally taxing.
- **Energy Consumption:** The energy required for training and running AI models can be substantial, contributing to a larger carbon footprint.

6. Security and Privacy Issues

- **Vulnerability to Attacks:** AI systems can be vulnerable to adversarial attacks where small, carefully crafted changes to input data can cause the AI to make incorrect decisions.
- **Privacy Concerns:** The use of AI in data analysis raises concerns about the privacy of individuals, as AI systems can infer sensitive information from seemingly innocuous data.

7. Human-AI Interaction

- **User Dependence:** Over-reliance on AI systems can lead to skill degradation in humans and an overdependence on automated systems.
- **Lack of Common Sense:** AI lacks the commonsense reasoning that humans possess, making it difficult for AI to handle unexpected situations or understand context as humans do.

8. Regulatory and Legal Challenges

- **Regulatory Uncertainty:** The fast-paced development of AI technologies often outstrips the creation of relevant regulations and laws, leading to uncertainty and potential misuse.
- **Legal Accountability:** Determining accountability and liability in cases where AI systems make erroneous or harmful decisions is complex and unresolved in many jurisdictions.

9. Economic and Social Impact

- **Job Displacement:** AI and automation can lead to job losses in certain sectors, contributing to economic disruption and social inequality.
- **Access and Equity:** The benefits of AI are not evenly distributed, with advanced AI technologies often being accessible only to large organizations or affluent regions, exacerbating existing disparities.

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Unit 2 : Logic for AI

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2. Predicate logic
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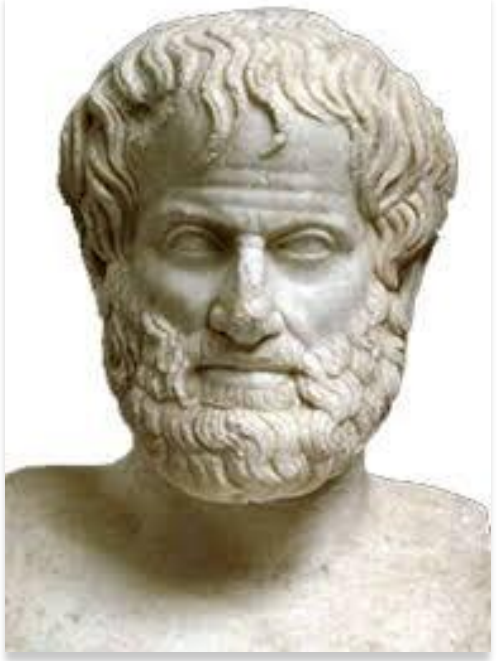
4. Clause form
5. Unification algorithm

Introduction to Logic

What is Logic?
The study of principles and methods used to distinguish **correct** reasoning from **incorrect** reasoning.

Father of Logic
Aristotle (384–322 BCE): Often considered the "father of logic," **Aristotle** developed the first formal system of logic, known as **syllogistic logic**. His work, particularly in "Organon," laid the groundwork for future logical studies.



Propositional Logic

- Propositional logic is also known as **propositional calculus** or **sentential logic**.
- It is a branch of logic that deals with **propositions** and their relationships through logical **connectives**.
- It forms the foundation for many areas of formal logic, computer science, and mathematics.

Propositions

- A proposition is a declarative statement that is either **True** or **False** but not both.
- Examples include:
  - "It is raining" (True or False)
  - "2 + 2 = 4" (True).

Logical Connectives

Connectives: Symbols representing logical operations.

Types of Connectives:

- *Negation (¬) :* The negation of a proposition P is denoted as ¬P and is True if P is False, and False if P is True.
- *Conjunction (∧) :* The conjunction of propositions P and Q is denoted as P∧Q and is True if both P and Q are True.
- *Disjunction (∨) :* The disjunction of propositions P and Q is denoted as P∨Q and is True any one of P or Q is True.
- *Implication (→) :* The implication P→Q is true if whenever P is True, Q is also True. It is False only if P is True and Q is False.
- *Bi-implication (↔) :* The biconditional P↔Q is true if P and Q are either both True or both False.

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Truth Tables

Purpose: Systematically determine the truth value of compound propositions.

Construction: Assign truth values to variables and apply logical connectives.

Example: Truth table for $(p \wedge q) \vee \neg p$

Truth Table Examples

P	$\neg P$
T	F
F	T

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

P	Q	$P \leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

For a detailed tutorial on Truth Tables, please refer to:
https://sites.millersville.edu/bikenaga/math-proof/truth-tables/truth-tables.html

Laws of Propositional Logic

- **Commutative Laws**: $p \wedge q \equiv q \wedge p$
- **Associative Laws**: $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
- **Distributive Laws**: $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
- **De Morgan's Laws**: $\neg(p \wedge q) \equiv \neg p \vee \neg q$
- **Idempotent Laws**: $p \wedge p \equiv p$

Example: Applying De Morgan's Law: $\neg(p \vee q) \equiv \neg p \wedge \neg q$

Examples

- Simplify the expression $\neg(p \vee q) \wedge (p \rightarrow q)$

Solution

Simplifying the expression $\neg(p \vee q) \wedge (p \rightarrow q)$

1. Apply De Morgan's Law to $\neg(p \vee q)$:

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

2. Apply the definition of implication to $p \rightarrow q$:

$$p \rightarrow q \equiv \neg p \vee q$$

3. Substitute these into the original expression:

$$(\neg p \wedge \neg q) \wedge (\neg p \vee q)$$

4. Distribute $\neg p$ across $(\neg p \wedge \neg q)$ and $(\neg p \vee q)$:

$$(\neg p \wedge \neg q) \wedge (\neg p \vee q) \equiv (\neg p \wedge \neg q \wedge \neg p) \vee (\neg p \wedge \neg q \wedge q)$$

5. Simplify the expression:

$$(\neg p \wedge \neg q \wedge \neg p) \vee (\neg p \wedge \neg q \wedge q) \equiv \neg p \wedge \neg q$$

So, the simplified expression is:

$$\neg p \wedge \neg q$$

What is Predicate Logic?

Predicate Logic: Deals with predicates and quantified variables.

Predicates: Statements containing variables that can be either true or false.

Quantifiers: Symbols indicating the extent of the variable's applicability (e.g., ∀ for universal quantification, ∃ for existential quantification).

Example: $\forall x(P(x))$ represents "for all x, P(x)".

Predicates and Quantifiers

Predicates: Expressions containing variables that can be true or false.

Quantifiers: Determine the scope of variables.

Examples:

Predicate: $P(x)$ - "x is a prime number."

Quantified Statement: $\forall x(P(x))$ - "All x are prime numbers."

Quantifier Negation

Negating Quantified Statements:

Universal Quantification: $\neg(\forall x(P(x))) \equiv \exists x(\neg P(x))$

Existential Quantification: $\neg(\exists x(P(x))) \equiv \forall x(\neg P(x))$

Example:

Negating "All cats are mammals"

becomes

"Some cats are not mammals".

Rules of Inference

- **Modus Ponens**: If p implies q and p is true, then q is true.
- **Modus Tollens**: If p implies q and q is false, then p is false.
- **Hypothetical Syllogism**: If p implies q, and q implies r, then p implies r.
- **Disjunctive Syllogism**: If p or q is true, and $\neg p$ is true, then q is true.
- **Constructive Dilemma**: If p implies q, and r implies s, and $p \vee r$ is true, then $q \vee s$ is true.

Example: Using Modus Ponens: If "It is raining" implies "The ground is wet", and "It is raining", then "The ground is wet".

Predicate Logic Inference Rules

- **Universal Instantiation**: If $\forall x(P(x))$ is true, then $P(a)$ is true for any individual a.
- **Existential Instantiation**: If $\exists x(P(x))$ is true, then $P(a)$ is true for some particular a.
- **Universal Generalization**: If $P(a)$ is true for any individual a, then $\forall x(P(x))$ is true.
- **Existential Generalization**: If $P(a)$ is true for some particular a, then $\exists x(P(x))$ is true.

Example: Using Universal Instantiation: If "All cats are mammals", then "Fluffy is a mammal".

Predicate Logic Equivalences

- **Universal Quantifier Equivalence**:

  $\neg\forall x(P(x)) \equiv \exists x(\neg P(x))$

- **Existential Quantifier Equivalence**:

  $\neg\exists x(P(x)) \equiv \forall x(\neg P(x))$

- **Negation Equivalence**:

  $\neg(\forall x(P(x))) \equiv \exists x(\neg P(x))$ and $\neg(\exists x(P(x))) \equiv \forall x(\neg P(x))$

Applications of Logic

Mathematics: Proving theorems, solving equations.

Computer Science: Designing algorithms, programming languages.

Philosophy: Analyzing arguments, studying language.

Linguistics: Understanding sentence structure, semantics.

Example: Using logic to verify software correctness.

What is Resolution?

- Resolution Method is an inference rule which is used in both proposition as well as First order predicate logic (FOPL) in different ways.
- This method is basically used for proving the satisfiability of a sentence.
- In resolution method, we use "**Proof by Refutation**" technique to prove the given statement.

What is Resolution?

- The Key idea for the resolution method is to use the knowledge base and negated goal to obtain null clause (which indicates contradiction).

- Resolution method is also called **Proof by Refutation**. Since the knowledge base itself is consistent, the contradiction must be introduced by a negated goal.

- As a result , we have to conclude that the original goal is true.

Resolution Method in AI

- There are two methods for resolution in AI:
  - **Resolution Method in Proposition logic**
  - **Resolution Method in Predicate logic**

Resolution Method in Propositional Logic

- In propositional logic, the resolution method is only the reference rule which gives a new clause when **two or more clauses** are coupled together.

- Using propositional logic, it becomes easy to make a theorem prover sound and complete for all.

Resolution Method in Propositional Logic

- The process followed to convert the propositional logic into resolution method contains the below steps:
  1. Convert the given axiom into clause form, i.e., Disjunction form.
  2. Apply and proof the given goal using negotiation rule.
  3. Using those literals which are needed to prove.
  4. Solve the clause together and achieve the goal.

Conjunctive Normal Form(CNF)

- In propositional logic, the resolution method is applied only to those clauses which are disjunction of literals. **There are following steps used to convert into CNF:**

1. Eliminate bi-conditional implication by replacing $A \Leftrightarrow B$ with $(A \rightarrow B) \wedge (B \rightarrow A)$
2. Eliminate implication by replacing $A \rightarrow B$ with $\neg A \vee B$.
3. In CNF, negation($\neg$) appears only in literals, therefore we move it inwards as:
   - $\neg(\neg A) \equiv A$ (double-negation elimination
   - $\neg(A \wedge B) \equiv (\neg A \vee \neg B)$ (De Morgan)
   - $\neg(A \vee B) \equiv (\neg A \wedge \neg B)$ (De Morgan)

Conjunctive Normal Form(CNF)

- Finally, using distributive law on the sentences, and form the CNF as:

$(A_1 \vee B_1) \wedge (A_2 \vee B_2) \wedge \ldots \wedge (A_n \vee B_n)$.

Note: CNF can also be described as AND of ORS

Disjunctive Normal Form (DNF)

- This is a reverse approach of CNF. The process is similar to CNF with the following difference:

$(A_1 \wedge B_1) \vee (A_2 \wedge B_2) \vee \ldots \vee (A_n \wedge B_n)$.

In DNF, it is OR of ANDS, a sum of products, or a cluster concept, whereas, in CNF, it is ANDs of Ors.

Example OF Propositional Resolution

- Consider the following Knowledge Base:

1. The humidity is high or the sky is cloudy.
2. If the sky is cloudy, then it will rain.
3. If the humidity is high, then it is hot.
4. It is not hot.

- **Goal:** It will rain.

- Use propositional logic and apply resolution method to prove that the goal is derivable from the given knowledge base.

Solution:

- Let's construct propositions of the given sentences one by one:
- The humidity is high or the sky is cloudy. ….. 1

  Let, P: Humidity is high.

  Q: Sky is cloudy.

  It will be represented as **P V Q.**
- If the sky is cloudy, then it will rain. …. 2

  Q: Sky is cloudy. …**from(1)**

  Let, R: It will rain.

  It will be represented as **Q → R.**

- If the humidity is high, then it is hot. …. 3

  P: Humidity is high. …**from(1)**

  Let, S: It is hot.

  It will be represented as **P** $\rightarrow$ **S.**

- It is not hot. …. 4

  ¬S: It is not hot.

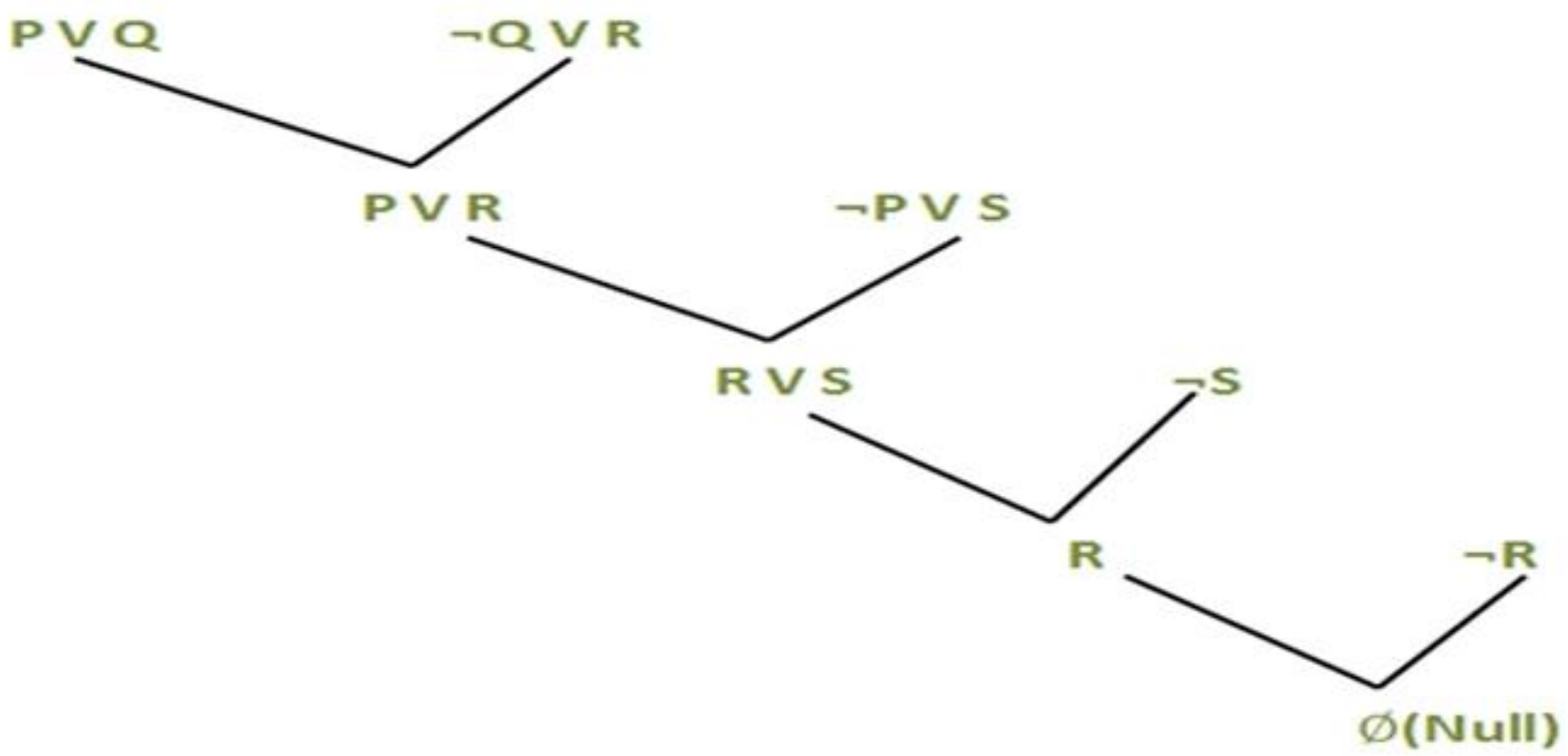
Applying resolution method

- Convert the given axiom into clausal form, i.e., disjunction form.
- Apply and proof the given goal using negation rule.
- Use those literals which are needed to prove.
- Solve the clauses together and achieve the goal.

- In (2), $Q \rightarrow R$ will be converted as $(\neg Q \vee R)$
- In (3), $P \rightarrow S$ will be converted as $(\neg P \vee S)$

- **Negation of Goal ($\neg R$):** It will not rain.

Finally, apply the rule as shown below:



After applying Proof by Refutation (Contradiction) on the goal, the problem is solved, and it has terminated with a **Null clause (Ø).** Hence, the goal is achieved. Thus, It is not raining.

By Mukesh Barapatre

Example:

Proof with Resolution

Given the following hypotheses:

1. If it rains, Joe brings his umbrella (r -> u)
2. If Joe has an umbrella, he doesn't get wet (u -> NOT w)
3. If it doesn't rain, Joe doesn't get wet (NOT r -> NOT w)

prove that Joes doesn't get wet (NOT w)

We first put each hypothesis in CNF:

1. r -> u == (NOT r OR u)
2. u -> NOT w == (NOT u OR NOT w)
3. NOT r -> NOT w == (r OR NOT w)

We then use resolution on the hypotheses to derive the conclusion (NOT w):

```
1. NOT r OR u          Premise

2. NOT u OR NOT w      Premise

3. r OR NOT w          Premise

4. NOT r OR NOT w      L1, L2, resolution

5. NOT w OR NOT w      L3, L4, resolution

6. NOT w               L5, idempotence

7. QED
```

Resolution Method in AI

- Resolution method is an inference rule which is used in both Propositional as well as First-order Predicate Logic in different ways.
- This method is basically used for proving the satisfiability of a sentence.
- In resolution method, we use **Proof by Refutation** technique to prove the given statement.

Resolution Method in AI

- The key idea for the resolution method is to use the knowledge base and negated goal to obtain null clause(which indicates contradiction).

- Resolution method is also called **Proof by Refutation**. Since the knowledge base itself is consistent, the contradiction must be introduced by a negated goal.

- As a result, we have to conclude that the original goal is true.

Resolution Method in FOPl/Predicate Logic

- Resolution method in FOPL is an uplifted version of propositional resolution method.
- **In FOPL, the process to apply the resolution method is as follows:**
  - Conversion of facts into first-order logic.
  - Convert FOL statements into CNF
  - Negate the statement which needs to prove (proof by contradiction)
  - Draw resolution graph (unification).

Example

1. John likes all kind of food.
2. Apple and vegetable are food
3. Anything anyone eats and not killed is food.
4. Anil eats peanuts and still alive
5. Harry eats everything that Anil eats.

Prove by resolution that:

John likes peanuts.

Step-1: Conversion of Facts into FOL

a) John likes all kind of food.

b) Apple and vegetable are food

c) Anything anyone eats and not killed is food.

d) Anil eats peanuts and still alive

e) Harry eats everything that Anil eats.

Prove by resolution that:

John likes peanuts.

a. $\forall x$: food(x) → likes(John, x)

b. food(Apple) ∧ food(vegetables)

c. $\forall x\ \forall y$: eats(x, y) ∧ ¬ killed(x) → food(y)

d. eats (Anil, Peanuts) ∧ alive(Anil).

e. $\forall x$: eats(Anil, x) → eats(Harry, x)

f. $\forall x$: ¬ killed(x) → alive(x)

g. $\forall x$: alive(x) → ¬ killed(x)

} added predicates.

h. likes(John, Peanuts)

Step-2: Conversion of FOL into CNF

1. Eliminate all implication (→) and rewrite
2. Move negation (¬)inwards and rewrite
3. Rename variables or standardize variables to unique variable.
4. Move all quantifiers to the left without changing their relative order.
5. Eliminate existential instantiation quantifier by elimination ∃ (Skolemization).
6. Drop Universal quantifiers ∀.
7. Convert the formula into a conjunction of disjuncts.

Step-2: Conversion of FOL into CNF

Eliminate all implication (→) and rewrite : A → B with ¬A V B.

a. ∀x: food(x) → likes(John, x)

b. food(Apple) Λ food(vegetables)

c. ∀x ∀y: eats(x, y) Λ ¬ killed(x) → food(y)

d. eats (Anil, Peanuts) Λ alive(Anil).

e. ∀x : eats(Anil, x) → eats(Harry, x)

f. ∀x: ¬ killed(x) → alive(x) } **added predicates.**

g. ∀x: alive(x) → ¬ killed(x) }

h. likes(John, Peanuts)

- ∀x ¬ food(x) V likes(John, x)
- food(Apple) Λ food(vegetables)
- ∀x ∀y ¬ [eats(x, y) Λ ¬ killed(x)] V food(y)
- eats (Anil, Peanuts) Λ alive(Anil)
- ∀x ¬ eats(Anil, x) V eats(Harry, x)
- ∀x¬ [¬ killed(x)] V alive(x)
- ∀x ¬ alive(x) V ¬ killed(x)
- likes(John, Peanuts).

Step-2: Conversion of FOL into CNF

Move negation (¬) inwards and rewrite

- ∀x ¬ food(x) V likes(John, x)
- food(Apple) Λ food(vegetables)
- **∀x ∀y ¬ [eats(x, y) Λ ¬ killed(x)] V food(y)**
- eats (Anil, Peanuts) Λ alive(Anil)
- ∀x ¬ eats(Anil, x) V eats(Harry, x)
- **∀x¬ [¬ killed(x)] V alive(x)**
- ∀x ¬ alive(x) V ¬ killed(x)
- likes(John, Peanuts).

- ∀x ¬ food(x) V likes(John, x)
- food(Apple) Λ food(vegetables)
- **∀x ∀y ¬ eats(x, y) V killed(x) V food(y)**
- eats (Anil, Peanuts) Λ alive(Anil)
- ∀x ¬ eats(Anil, x) V eats(Harry, x)
- **∀x killed(x)] V alive(x)**
- ∀x ¬ alive(x) V ¬ killed(x)
- likes(John, Peanuts).

Step-2: Conversion of FOL into CNF

Rename variables or standardize variables

- ∀x ¬ food(x) V likes(John, x)
- food(Apple) Λ food(vegetables)
- ∀x ∀y ¬ eats(x, y) V killed(x) V food(y)
- eats (Anil, Peanuts) Λ alive(Anil)
- ∀x ¬ eats(Anil, x) V eats(Harry, x)
- ∀x ¬killed(x)] V alive(x)
- ∀x ¬ alive(x) V ¬ killed(x)
- likes(John, Peanuts).

- ∀x ¬ food(x) V likes(John, x)
- food(Apple) Λ food(vegetables)
- ∀y ∀z ¬ eats(y, z) V killed(y) V food(z)
- eats (Anil, Peanuts) Λ alive(Anil)
- ∀w¬ eats(Anil, w) V eats(Harry, w)
- ∀g ¬killed(g)] V alive(g)
- ∀k ¬ alive(k) V ¬ killed(k)
- likes(John, Peanuts).

Step-2: Conversion of FOL into CNF

Eliminate existential instantiation quantifier by elimination.

- $\forall x \neg food(x) \vee likes(John, x)$
- $food(Apple) \wedge food(vegetables)$
- $\forall y \forall z \neg eats(y, z) \vee killed(y) \vee food(z)$
- $eats(Anil, Peanuts) \wedge alive(Anil)$
- $\forall w \neg eats(Anil, w) \vee eats(Harry, w)$
- $\forall g \neg killed(g)] \vee alive(g)$
- $\forall k \neg alive(k) \vee \neg killed(k)$
- $likes(John, Peanuts)$.

Step-2: Conversion of FOL into CNF

Drop Universal quantifiers.

- ∀x ¬ food(x) V likes(John, x)
- food(Apple) Λ food(vegetables)
- ∀y ∀z ¬ eats(y, z) V killed(y) V food(z)
- eats (Anil, Peanuts) Λ alive(Anil)
- ∀w¬ eats(Anil, w) V eats(Harry, w)
- ∀g ¬killed(g)] V alive(g)
- ∀k ¬ alive(k) V ¬ killed(k)
- likes(John, Peanuts).

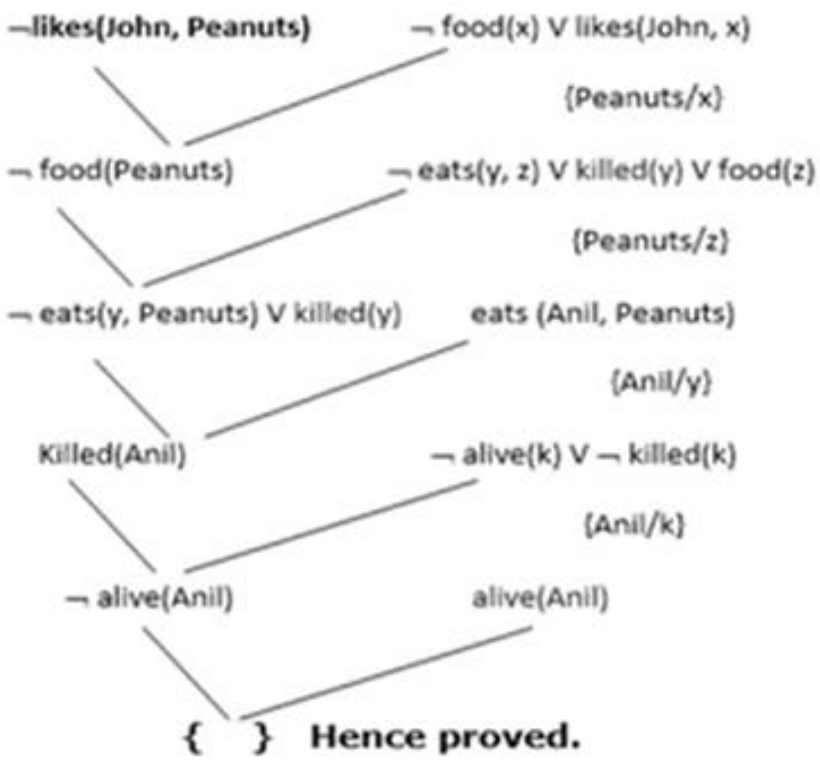
- ¬ food(x) V likes(John, x)
- food(Apple)
- food(vegetables)
- ¬ eats(y, z) V killed(y) V food(z)
- eats (Anil, Peanuts)
- alive(Anil)
- ¬ eats(Anil, w) V eats(Harry, w)
- killed(g) V alive(g)
- ¬ alive(k) V ¬ killed(k)
- likes(John, Peanuts).

Step-3: Negate the statement to be proved

- In this statement, we will apply negation to the conclusion statements, which will be written as

¬likes(John, Peanuts)

Draw Resolution graph



- ¬ food(x) V likes(John, x)
- food(Apple)
- food(vegetables)
- ¬ eats(y, z) V killed(y) V food(z)
- eats (Anil, Peanuts)
- alive(Anil)
- ¬ eats(Anil, w) V eats(Harry, w)
- killed(g) V alive(g)
- ¬ alive(k) V ¬ killed(k)
- likes(John, Peanuts).

https://www.javatpoint.com/ai-resolution-in-first-order-logic

https://www.youtube.com/watch?v=CbI-Q2a5rUU

Thank You